HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Python** automation scripts to streamline network configuration validation and compliance checks across thousands of global sites.
- Developed automated remediation workflows using **Python** to resolve common network issues without human intervention, improving system resilience.
- Contributed to an internal **Python**-based data pipeline for analyzing network performance patterns across distributed infrastructure, enabling proactive issue identification.
- Engineered **Java Spring Boot** microservices for internal tooling, focusing on robust REST API design and efficient data exchange with other backend systems.
- Authored **Bash** shell scripts for log analysis and system monitoring tasks on Linux servers, aiding in rapid troubleshooting of production incidents.
- Containerized network automation tools using **Docker** and deployed them to an **AWS** environment, ensuring consistent execution across diverse infrastructure.
- Collaborated with firmware and field operations engineers to conduct technical reviews and validate new hardware configurations before production rollout.

SKILLS

Networking – Wireless (Wi-Fi 5/6/6E/7), Wired (802.1X, NAC), IP routing protocols, TCP/IP, OSPF/IS-IS, BGP, MPLS, Server Load Balancers, Firewalls, ACLs, DNS, DHCP, IPAM, RADIUS

Automation & Scripting – Python, Bash, Shell scripting, Perl, PHP, Ruby

Cloud & Containerization – AWS, EC2, S3, Lambda, CloudWatch, Docker, Kubernetes

Programming Languages – Java, Spring Boot, Python, Perl, PHP, Ruby, Bash, Shell

Development Practices – Agile, Scrum, CI/CD, Jenkins, Git, Code Reviews

PROJECTS

Global Wireless Network Automation Framework

- Developed a **Python** framework for zero-touch deployment of wireless network devices across thousands of global sites, significantly reducing manual configuration effort.
- Integrated **Bash** scripting to automate pre-deployment hardware validation checks, catching configuration drift and firmware incompatibilities prior to network integration.
- Designed and implemented automated remediation workflows using **Python** and AWS Lambda to self-heal common network connectivity issues, reducing incident response time by an average of 45%.

Network Anomaly Detection and Alerting System

- Engineered a predictive maintenance capability by developing anomaly detection models in **Python** (scikit-learn) on time-series network performance data.
- Built data pipelines using **Python** and SQL to aggregate and process metrics from distributed network sensors, feeding into the anomaly detection models.
- Created near real-time dashboards using **Power BI** to visualize network health, detected anomalies, and automated alerts, providing operational teams with actionable insights.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)